

12.2 Stationary Waves

1 C

2 B

3 D

4 C

5 C

6 C

7 A

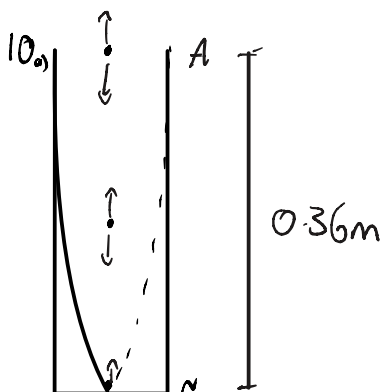
8 $v = f\lambda$

$$f = \frac{v}{\lambda} \quad \lambda = 2x$$

$$f = \frac{c}{2x} \quad \underline{C}$$

9 a) waves travelling in opposite directions are superposed
same frequency, velocity, & wavelength
similar amplitudes

b) air molecules on either side of a node are equal in magnitude but opposite in direction
 $\therefore$ interference occurs
Pressure at a node



b) $0.36\text{m} = \frac{1}{4}\lambda$
 $\lambda = 1.44\text{m}$

c) $v = 330$
 $\lambda = 1.44$

$v = f\lambda$

$f = \frac{v}{\lambda} = \frac{330}{1.44} = 230 \text{ Hz}$

d) open @ both ends, 1st series harmonic

$\rightarrow \frac{1}{2}\lambda \text{ of wave } (1\lambda)$

$\frac{1}{2}\lambda = 0.72 \text{ m}$

$f = \frac{v}{\lambda} = \frac{330}{0.72} = 460 \text{ Hz}$

11 a) A stationary wave oscillates in time and does not propagate.

A stationary wave also has nodes & antinodes

b) The distance between the clamp and the pulley is $\frac{n}{2}\lambda$ apart, where $n \in \mathbb{N}$ (n is an integer). So in stationary wave terms (in this example, $80 \text{ cm} = \frac{n}{2}\lambda = 1\lambda$).

The clamp initially oscillates at some frequency f .

As the distance between the clamp and the pulley is 1λ , the reflected wave = the non-reflected wave, but 180° out of phase.

So a stationary wave forms. (The waves are superposed).

c) i) P and Q are equidistant from the antinode (located 20 cm from the clamp). So amplitudes are the same at any time. All points oscillate at the same frequency, as the wave is stationary.

Because P and S (and Q and S) are opposite sides of the central node they oscillate opposite to each other (e.g. when P/Q is at its peak, S is at a trough, and vice versa).

Also, the amplitude of S > the amplitude of P/Q unless $|S|$ (here, the amplitude of S) = 0. Then, $|S| = |P|/|Q|$.

d) i) To be in phase, $\lambda = \frac{1}{8} \times 80 \text{ cm} = 10 \text{ cm}$
 v is constant $\therefore f = 8 \times 30 = 240 \text{ Hz}$.

ii) 0 Hz , because then the wave is not moving $\therefore$ all points are at rest.

$$12 f = 500 \text{ Hz.}$$

$$\text{Boundary conditions } l_{\text{wire}} = \frac{2n-1}{4} \lambda, \quad n \in \mathbb{N}^+$$

$$\text{Let } l_1 = \frac{2n-1}{4} \lambda \text{ where } n = n^{\text{th}} \text{ harmonic.}$$

$$\text{Then } l_2 = \frac{2(n-1)-1}{4} \lambda \text{ where } n-1 = (n-1)^{\text{th}} \text{ harmonic}$$

$$= \frac{2n-3}{4} \lambda$$

$$l_1 - l_2 = \frac{2n-1-(2n-3)}{4} \lambda = \frac{2}{4} = \frac{1}{2} \lambda$$

$$= 0.336 \text{ m}$$

$$\therefore \lambda = 0.672 \text{ m.}$$